



Life-sized Gundam robot

Japanese engineers built a life-sized mecha from the Gundam series, which recently took its first steps. <http://digit.in/oct20-23>



Amazon announces Luna

Amazon jumped into cloud gaming by launching Luna, a video game streaming service. <http://digit.in/oct20-24>

them. This is the area where most of the TTS systems fail to deliver and they can only be evaluated using different synthesizers with a certain text being read out to different volunteers to get a clear understanding of the comprehensibility of the TTS.

EVOLUTION OF SMOOTH TALKING IN AI

The first speech synthesis experiment was conducted in 1779 by the Russian Professor Christian Kratzenstein who created an apparatus based on the human vocal tract to depict the 'physiological differences involved in the production of five long vowel sounds'. This was the experiment which created a foundation for including speech synthesis in AI. The second advancement to the Kranztein experiment was in 1952 when Bell laboratories created the "Audrey" system which could

in 1970, the speech understanding research program (SUR) was conducted using the Carnegie Mellon made "Harpy" speech system and the system was able to understand and replicate 1000 words. Also, voice interpretation was made very common in the system which was again the foundation for the AI that came to be in the future.

After that the HMM (Hidden Markov Model) was postulated which drastically increased the vocabulary understood by machines from a few hundred to the thousands. Also the HMM started pronouncing words which weren't actually actual words which was a huge advancement.

The speech recognition and speech synthesis processes skyrocketed in the 90s considering the usage of fast processors compared to what they had earlier. Bellsouth

In the 2000s, Google announced the Google voice and following this the accuracy of the speech recognition and synthesis was around 80 percent. It was also distributed to a lot of people in the form of applications and people actually started using this and the AI used started predicting the speech patterns of the users. The vocabulary of the system had drastically increased from a few hundreds to billions.

In the later 2010s, Apple introduced the Siri voice assistant and following this the voice assistants being introduced started booming and this was the starting point for a new AI speech dependent era. Today, some of the largest tech companies are competing to herald the speech accuracy title. In 2016, IBM achieved a word error rate of 6.9 percent. In 2017 Microsoft usurped IBM with a 5.9 percent claim. Shortly after that IBM improved their rate to 5.5 percent. However, it is Google that is claiming the lowest rate of errors when it comes to voice assistants at a mere 4.9 percent, which is impressive.

HOW THESE SPEECH CAPABILITIES COULD BE MISUSED?

The path to smooth talking AI has to come with certain sacrifices but the question is, should privacy be included as one of the sacrifices. In order to increase the effectiveness of the AI, adequate data should be given to the machines for predictions and speech pattern analysis. And before actually accessing the data the AI asks for a bleak consent form which clearly does not include the actual usages of the data being given to the machines.

The data protection model today in India is consent-based and under the consent-based system, the data user or the data manipulator is free to use and process the data from the user once the consent is obtained from the user. The user also isn't fully made aware as to what the



<https://unsplash.com>

recognise a single voice speaking digits aloud and shortly after that IBM introduced another AI which could understand and respond to 16 words in English which was astounding enhancement of the primary idea considering the primitive backgrounds. Shortly after that

introduced the voice portal (VAL) which was basically a dial in interactive speech recognition system. The system used in this interactive experiment gave rise to a lot of phone tree systems in existence today. Which is certainly an important part of our lives.